Invariance: Problem Session

STAT 220B – Working the proof techniques

How to use this session

These problems are not exam problems. Each one isolates one proof technique you will need for invariance questions. We have already worked HW4 Problem 1 (Pitman for the Uniform, giving the midrange); today we drill the surrounding machinery on fresh models.

The five techniques, one per problem:

1. **Derive the equivariant class** from the functional equation $\delta(gx) = \tilde{g}\,\delta(x)$.
2. **Constant risk on orbits** (Theorem 1), then minimize the constant: two equivariant rules can both have constant risk, and you still have to optimize.
3. **Pitman as generalized Bayes** under the flat prior, for a non-normal location family.
4. **Best equivariant for a scale family** via the right-invariant (Haar) prior $1/\sigma$, cross-checked against direct minimization.
5. **Hunt-Stein and the admissibility tension**: when does best equivariant equal minimax, and why does James-Stein not contradict it.

Suggested pacing: about 15 minutes each, with the last one shorter and discussion-driven.

Problem 1: Reading the equivariant class off the functional equation

The problem

Let $X_1, \dots, X_n$ be iid from a scale family $f(x \mid \sigma) = \sigma^{-1} f_0(x/\sigma)$, $\sigma > 0$, where f_0 is a fixed density on $(0, \infty)$. The scale group is $g_b(x) = bx$, $b > 0$, with induced actions $\bar{g}_b(\sigma) = b\sigma$ and $\tilde{g}_b(a) = ba$. Let $T = T(X_1, \dots, X_n)$ be any statistic satisfying $T(bx_1, \dots, bx_n) = b\,T(x_1, \dots, x_n)$ (scale-equivariant), with $T > 0$ almost surely.

(a) Write down what equivariance $\delta(g_b \mathbf{x}) = \tilde{g}_b \delta(\mathbf{x})$ requires of a rule δ built from T, i.e. of the form $\delta(\mathbf{x}) = h(T(\mathbf{x}))$.

(b) By choosing b cleverly, show that equivariance forces $h(t) = c\,t$ for a constant $c > 0$. Conclude that the equivariant rules based on T form the one-parameter family $\{cT : c > 0\}$.

(c) State the analogous conclusion for a **location** family with T location-equivariant ($T(\mathbf{x} + d\mathbf{1}) = T(\mathbf{x}) + d$): what one-parameter family do you get?

Technique: the "set b to kill the data" move

This is the workhorse first step of almost every invariance computation.

(a) Equivariance reads $h(T(b\mathbf{x})) = b\,h(T(\mathbf{x}))$. Since T is scale-equivariant, $T(b\mathbf{x}) = bT(\mathbf{x})$, so the condition becomes

$$h(b\,T(\mathbf{x})) = b\,h(T(\mathbf{x})) \qquad \text{for all } b > 0, \text{ all } \mathbf{x}.$$

(b) Write $t = T(\mathbf{x}) > 0$. The condition $h(bt) = b\,h(t)$ holds for all $b > 0$. Set $b = 1/t$:

$$h(1) = \frac{1}{t}\,h(t) \quad \Longrightarrow \quad h(t) = h(1)\,t = c\,t, \qquad c := h(1) > 0.$$

So every equivariant rule based on T is $\delta_c(\mathbf{x}) = cT(\mathbf{x})$. The infinite-dimensional rule space has collapsed to one positive constant.

(c) For a location family the same move on $h(T(\mathbf{x}) + d) = h(T(\mathbf{x})) + d$ (set $d = -T(\mathbf{x})$) gives $h(t) = t + k$ with $k = h(0)$. The equivariant family is $\{T + k : k \in \mathbb{R}\}$.

Teaching point. The choice of b (or d) that cancels the data is the whole trick: $b = 1/t$ for scale, $d = -t$ for location. Memorize the move, not the answer. Everything downstream (find the best c or k) is then a one-variable calculus problem.

Problem 2: Two equivariant rules, both constant risk

The problem

Let $X_1, \dots, X_n$ be iid from the shifted-exponential location family $f(x \mid \theta) = e^{-(x-\theta)}$ for $x > \theta$, $\theta \in \mathbb{R}$, under squared-error loss. (This is the same family as one of the practice exam problems; here the point is to compare two equivariant rules, not to find the single best one.) Consider

$$\delta_A(\mathbf{X}) = \bar{X} - 1, \qquad \delta_B(\mathbf{X}) = X_{(1)} - \tfrac{1}{n}.$$

(a) Verify both are location-equivariant.
(b) Show each has constant risk in θ (cite the right theorem), and compute both risks.
(c) Which is better, and by how much as $n \to \infty$? What is the moral about equivariance and optimality?

Technique: Theorem 1 plus a one-line risk each

(a) Under $\mathbf{x} \mapsto \mathbf{x} + d\mathbf{1}$: $\bar{X} \mapsto \bar{X} + d$ so $\delta_A \mapsto \delta_A + d$; and $X_{(1)} \mapsto X_{(1)} + d$ so $\delta_B \mapsto \delta_B + d$. Both equivariant.

(b) By Theorem 1 (Lecture 11), an equivariant rule has constant risk on the orbit of $\bar{\mathcal{G}}$, which for the location group acting on $\mathbb{R}$ is all of Θ. So both risks are constant in θ; compute at $\theta = 0$, i.e. with $X_i \sim \mathrm{Exp}(1)$ (mean 1, variance 1).

For δ_A: $\mathbb{E}_0[\bar{X}] = 1$, so $\bar{X} - 1$ is unbiased, and $R(\theta, \delta_A) = \mathrm{Var}(\bar{X}) = \frac{1}{n}$.

For δ_B: $X_{(1)} \sim \mathrm{Exp}(\text{rate } n)$ at $\theta = 0$ (minimum of n iid $\mathrm{Exp}(1)$), so $\mathbb{E}_0[X_{(1)}] = 1/n$, making $X_{(1)} - 1/n$ unbiased, and $R(\theta, \delta_B) = \mathrm{Var}(X_{(1)}) = \frac{1}{n^2}$.

(c) δ_B is better, with risk ratio $R_B/R_A = (1/n^2)/(1/n) = 1/n \to 0$: the minimum-based rule is asymptotically infinitely better than the mean-based rule.

Teaching point. Equivariance is a *constraint*, not an *optimization*. Both rules respect the symmetry and both have flat risk, but flatness does not mean optimality. The minimum carries far more location information than the mean for this one-sided family, so you still have to minimize over the equivariant family. Equivariance narrows the search; it does not finish it.

Problem 3: Pitman for a non-normal location family

The problem

Let $X_1, \dots, X_n$ be iid Laplace (double-exponential) with location θ:
$f(x \mid \theta) = \frac{1}{2} e^{-|x-\theta|}$, $\theta \in \mathbb{R}$, under squared-error loss.
(a) Write down the Pitman estimator as a generalized Bayes rule under the flat prior $\pi(\theta) = 1$.
(b) For $n = 2$ with order statistics $x_{(1)} \le x_{(2)}$, show that the posterior $\propto e^{-|x_1-u|-|x_2-u|}$ is symmetric about the midpoint $(x_{(1)} + x_{(2)})/2$, and hence that the Pitman estimator is that midpoint.
(c) For $n = 3$, explain (no closed form needed) why the Pitman estimator is neither the sample mean nor the sample median.

Technique: write the Bayes integral, exploit symmetry, recognize the obstruction

(a) The Pitman estimator is the posterior mean under $\pi(\theta) = 1$:

$$\delta^*(\mathbf{x}) = \frac{\displaystyle\int_{-\infty}^{\infty} u \prod_{i=1}^{n} f(x_i - u)\, du}{\displaystyle\int_{-\infty}^{\infty} \prod_{i=1}^{n} f(x_i - u)\, du} = \frac{\displaystyle\int u\, e^{-\sum_i |x_i - u|}\, du}{\displaystyle\int e^{-\sum_i |x_i - u|}\, du}.$$

(b) For $n = 2$, examine the exponent $-(|x_1 - u| + |x_2 - u|)$ as a function of u (take $x_1 = x_{(1)} \le x_{(2)} = x_2$):

$$\sum_i |x_i - u| = \begin{cases} (x_1 + x_2) - 2u, & u < x_{(1)}, \\ x_{(2)} - x_{(1)}, & x_{(1)} \le u \le x_{(2)} \text{ (constant)}, \\ 2u - (x_1 + x_2), & u > x_{(2)}. \end{cases}$$

So the unnormalized posterior $e^{-\sum_i |x_i - u|}$ is a flat plateau on $[x_{(1)}, x_{(2)}]$ with two symmetric exponential shoulders. The whole shape is symmetric about the midpoint $m = (x_{(1)} + x_{(2)})/2$: reflecting $u \mapsto 2m - u$ leaves it unchanged. A symmetric density has its mean at the center of symmetry, so $\delta^* = (x_{(1)} + x_{(2)})/2$.

Technique (continued):

(c) For $n = 3$ the median observation breaks the symmetry: the exponent $\sum_i |x_i - u|$ is piecewise linear with kinks at each $x_{(i)}$ and a unique minimum at the sample median (that is the MLE), but the posterior $e^{-\sum_i |x_i - u|}$ is asymmetric about that median unless the points happen to be symmetric. The posterior *mean* is pulled toward the longer tail, so it differs from the median; and because the shoulders are exponential rather than Gaussian, it also differs from the sample mean. The Pitman estimator is a genuine weighted integral with no elementary closed form.

Teaching point. Two transferable moves. First, the Pitman estimator is always the flat-prior posterior mean: write that integral immediately. Second, when the integrand is symmetric about a point, the mean is that point, no integration needed. When symmetry fails (generic $n \geq 3$), recognize that the answer is an honest integral and that the MLE (median here) and the sample mean are both wrong.

Problem 4: Best equivariant scale estimator two ways

The problem

Let $X \sim \mathrm{Exponential}$ with scale θ: $f(x \mid \theta) = \theta^{-1} e^{-x/\theta}$, $x > 0$, a single observation. Estimate θ under the scale-invariant loss $L(\theta, a) = (a/\theta - 1)^2$.

(a) **Direct route.** Using the equivariant family $\{cX : c > 0\}$ from Problem 1 and $X/\theta \sim \mathrm{Exp}(1)$ (so $\mathbb{E}[X/\theta] = 1$, $\mathbb{E}[(X/\theta)^2] = 2$), find the best equivariant estimator.

(b) **Haar-prior route.** The right-invariant prior for the scale group is $\pi(\theta) = 1/\theta$. Show the formal posterior given $X = x$ is inverse-gamma$(1, x)$, and using the inverse-gamma moments $\mathbb{E}[1/\theta \mid x] = 1/x$ and $\mathbb{E}[1/\theta^2 \mid x] = 2/x^2$, derive the generalized Bayes rule under L.

(c) Confirm the two routes give the same estimator. State the general principle this illustrates.

Technique: equivariant minimization and the Haar-prior Bayes rule must agree

(a) For $\delta_c = cX$, with $W = X/\theta \sim \mathrm{Exp}(1)$,

$$R(\theta, \delta_c) = \mathbb{E}[(cW - 1)^2] = c^2 \mathbb{E}[W^2] - 2c\,\mathbb{E}[W] + 1 = 2c^2 - 2c + 1,$$

constant in θ. Minimize: $\frac{d}{dc}(2c^2 - 2c + 1) = 4c - 2 = 0$, so $c^* = \frac{1}{2}$ and

$$\delta^*(X) = \tfrac{1}{2}X.$$

The second derivative $4 > 0$ confirms a minimum.

Technique (continued): the Haar route and why they agree

(b) The formal posterior is $\pi(\theta \mid x) \propto \theta^{-1} e^{-x/\theta} \cdot \theta^{-1} = \theta^{-2} e^{-x/\theta}$, the kernel of inverse-gamma$(1, x)$. Under $L(\theta, a) = a^2/\theta^2 - 2a/\theta + 1$, the Bayes action minimizes $a^2\, \mathbb{E}[1/\theta^2 \mid x] - 2a\, \mathbb{E}[1/\theta \mid x] + 1$, giving (set derivative to zero)

$$a^* = \frac{\mathbb{E}[1/\theta \mid x]}{\mathbb{E}[1/\theta^2 \mid x]} = \frac{1/x}{2/x^2} = \frac{x}{2}.$$

(c) Both routes give $\delta^*(X) = X/2$. This illustrates the general theorem (Lecture 11): for a scale family, the best scale-equivariant estimator equals the generalized Bayes rule under the right-invariant Haar prior $1/\sigma$. For location families the analogous Haar prior is the flat prior (Problem 3). Either route is valid; pick whichever is less work for the model in front of you. Here the direct route is faster, but the Haar route is the one that generalizes and explains *why* the equivariant answer is "Bayesian" at all.

Teaching point. If a problem says "best equivariant," you have two interchangeable tools: minimize the constant risk over $\{cT\}$, or compute the generalized Bayes rule under the Haar prior. On an exam, doing one and sanity-checking with the other catches arithmetic slips.

Problem 5: Hunt-Stein and the admissibility tension

The problem

This one is conceptual; argue in words and short formulas, no integrals.
(a) State the Hunt-Stein theorem in the form you would use it: under what condition on the group does "best equivariant" imply "minimax," and what are the two side hypotheses you must check?
(b) For estimating θ in $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$ under translation, the best location-equivariant estimator is $\bar{\mathbf{X}} = \mathbf{X}$. For $p \geq 3$ this is inadmissible (James-Stein dominates it). Explain precisely why this does **not** contradict Hunt-Stein.
(c) Name an amenable group and a non-amenable group, and state in one sentence what can go wrong with the Hunt-Stein conclusion when the group is non-amenable.

Technique: separate "minimax" from "admissible," and check amenability

(a) Hunt-Stein (Lecture 11): if the decision problem is invariant under an **amenable** group, then the best equivariant rule is minimax. The two side hypotheses to check are: (i) the loss is bounded below, and (ii) the risk function is suitably regular (finite, and continuous in θ). When these hold and the group is amenable, $\sup_\theta R(\theta, \delta^*_{\mathsf{equiv}}) = \inf_\delta \sup_\theta R(\theta, \delta)$.

(b) Hunt-Stein says $\mathbf{X}$ is *minimax*, and that remains true for every p: its constant risk p equals the minimax value. Hunt-Stein says nothing about *admissibility*. Minimax and admissible are different properties: a rule can be minimax (best worst-case) yet still be uniformly beaten by another rule that is also minimax. For $p \geq 3$, James-Stein has $R(\theta, \delta^{JS}) < p$ for all θ, so it too is minimax (its sup risk is $\leq p$), and it dominates $\mathbf{X}$. Crucially, δ^{JS} is **not equivariant**: it shrinks toward a fixed point, which breaks translation symmetry, so it is not a competitor *within* the equivariant class that Hunt-Stein optimizes over. No contradiction: invariance pins down the best *equivariant* rule and certifies it minimax, but it offers no admissibility guarantee.

Technique (continued): amenability and the four labels

(c) Amenable: the location group $(\mathbb{R}, +)$, the scale group $(\mathbb{R}_+, \cdot)$, and location-scale; also compact groups. Non-amenable: the general linear group GL_p for $p \geq 2$ (and free groups). When the group is non-amenable, the averaging construction behind Hunt-Stein fails, and the best equivariant rule need not be minimax.

Teaching point. Keep four labels mentally separate: equivariant, minimax, admissible, Bayes. Hunt-Stein links the first two (under amenability). The Stein phenomenon shows the second and third can come apart. An exam question that asks "is the best equivariant rule minimax / admissible?" is really asking you to recite which links hold and which do not.

Wrap-up: the five techniques

Problem	Technique	The move to remember
1	Derive the equivariant class	Set $b = 1/T$ (scale) or $d = -T$ (location) to collapse h
2	Constant risk on orbits	Theorem 1 flattens risk; still minimize the constant
3	Pitman as flat-prior posterior mean	Write the integral; use symmetry; know when there is no closed form
4	Best equivariant scale, two ways	Minimize $\{cT\}$, or Haar prior $1/\sigma$; they agree
5	Hunt-Stein and admissibility	Amenable $\Rightarrow$ minimax; minimax $\neq$ admissible; JS is not equivariant

Wrap-up (continued): one principle behind all five

Across all five: invariance is a **reduction principle**. It uses the symmetry of the problem to shrink the candidate rules to a one-parameter family, after which an ordinary optimization (or a Haar-prior Bayes calculation) finishes the job. The recurring exam traps are (i) forgetting that you still have to minimize after imposing equivariance, and (ii) conflating "minimax" with "admissible."